Predicting School-Level ACT Scores Using Socioeconomic Factors: A Regression Analysis

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Abstract

This study investigates whether socioeconomic variables improve the prediction of school-level ACT scores beyond a baseline model using the percentage of students eligible for free or reduced-price lunch. Three datasets from EdGap, the National Center for Education Statistics (NCES), and Zillow were combined, resulting in 7,227 high schools after data cleaning and imputation. Five socioeconomic variables were evaluated using linear, logarithmic, and quadratic functional forms, with model performance assessed using R^2 and mean absolute error (MAE). Among the initial extended models, the percentage of adults with a college degree achieved the best performance (R^2 = 0.6332, MAE = 1.1245). Adding a quadratic term increased R^2 to 0.6407, while MAE remained unchanged. These results suggest that socioeconomic variables, particularly adult college attainment, can improve the explanation of variation in school-level ACT scores. However, geographic limitations and imputed housing prices may affect the generalizability of the findings.

1 Introduction

1.1 Background and Scientific Aim

Previous studies have examined the relationship between socioeconomic conditions and academic achievement. Research on the Community Eligibility Provision (CEP), a program that provides free meals to eligible schools, has found positive effects on student outcomes. For example, Gordanier et al. (2020), in Free Lunch for All!, found that CEP was associated with an approximately 0.06 standard deviation increase in mathematics test scores among elementary school students. However, the existing literature often focuses on specific programs or geographic areas, leaving room to investigate the predictive value of a broader range of socioeconomic characteristics across multiple states.

This study examines whether incorporating additional socioeconomic variables improves the prediction of school-level ACT scores beyond a baseline model using the percentage of students eligible for free or reduced-price lunch. Specifically, five variables are considered: unemployment rate, housing price, household income, the percentage of children living in married-couple families, and the percentage of adults with a college degree. The central research question is whether these additional socioeconomic variables improve the predictive performance of the baseline model.

1.2 Data Description

Three datasets are used in this project. The first is the EdGap dataset (EdGap) from EdGap.org. It includes average ACT scores for schools, using SAT-based scores where ACT scores were unavailable, and several socioeconomic characteristics of the school district from 2016. Key variables include unemployment rate, adults with a college degree, children in married-couple families, percentage of students at the school eligible for free or reduced-price lunch, and household income. The second dataset provides school-level information (School Info) from the National Center for Education Statistics (NCES). It includes basic school information such as address, school type, school level, and charter school indicator. The last dataset is housing price data containing monthly housing prices by ZIP code from Zillow.

The EdGap and NCES datasets were merged using a left join on school ID. Because ZIP codes were stored as integers in the source files, leading zeros were lost (e.g., 2108 instead of 02108), so all ZIP codes were converted to strings and zero-padded to five digits. Housing price data from June 30, 2016, were then joined by ZIP code to align with the year of the EdGap data. The merged data contained 7,986 rows. For rate-based

variables, values less than or equal to zero were set to NA because they are not plausible. Only high schools were selected because we focus on ACT scores, and rows with an average ACT score of less than 1 were deleted as invalid data. Rows missing average ACT scores were excluded because the response variable is required for modeling, leaving 7227 rows.

The 237 missing housing prices were imputed using the mean of surrounding values based on zip code. Other missing variables, including rate_unemployment (n=22), percent_college (n=11), percent_married (n=34), median_income (n=16), and percent_lunch (n= 67), were imputed with scikit-learn's `IterativeImputer`, which estimates missing value based on other available features (scikit-learn developers, n.d.).

2 Theoretical Background

2.1 Simple and Multiple Regression Models

Simple linear regression predicts a quantitative response variable (Y) using a single predictor variable (X), assuming an approximately linear relationship between them (James et al., 2023). Multiple linear regression extends this approach by incorporating multiple predictor variables to explain or predict the response variable.

The relationship between a predictor and a response variable does not always follow a straight line. Therefore, regression models can incorporate different functional forms to capture nonlinear relationships. This study considers three functional forms: linear, logarithmic, and quadratic. A linear model assumes a constant change in the response for a one-unit increase in the predictor. A logarithmic model applies a log transformation to the predictor to capture relationships in which the response changes with the relative rather than absolute increase in the predictor. A quadratic model includes both the predictor and its squared term to capture curvature in the relationship.

2.2 Functional Forms

Log transformation is often applied to right-skewed data because extreme values can disproportionately influence a regression model and make it difficult to capture the underlying relationship between variables. By compressing large values more than small values, log transformation can reduce skewness and help linearize nonlinear relationships.

Quadratic transformation is used when the relationship between a predictor and the response variable is nonlinear and exhibits curvature. By including both the predictor and its squared term in the regression model, it can capture patterns such as diminishing or increasing effects, as well as U-shaped or inverted U-shaped relationships. This allows the model to represent more complex relationships than a simple linear model.

2.3 Evaluation Metrics

This study evaluates model performance using the coefficient of determination(R^2) and mean absolute error (MAE). The $\boldsymbol{R^2}$ measures the proportion of variation in the response variable explained by the model, with higher values generally indicating better predictive performance. **MAE** measures the average absolute difference between the observed and predicted ACT scores, with lower values indicating more accurate predictions.

3 Methodology

First, a baseline regression model is constructed using the percentage of students eligible for free or reduced-price lunch to predict school-level ACT scores. Next, five additional socioeconomic variables are evaluated using three functional forms: linear, logarithmic, and quadratic. For each variable, the functional form with the best performance is selected based on the model evaluation results.

Each selected variable is then added individually to the baseline model to construct an extended model. The baseline and extended models are evaluated using the$\boldsymbol{R^2}$ and MAE. An extended model is considered to improve predictive performance only if it achieves both a

higher R^2 and a lower MAE than the baseline model. If only one metric improves, the extended model is not considered better under this criterion.

Next, scatter plots and regression curves are examined to visualize the relationships between the socioeconomic variables and ACT scores and to identify potential opportunities for model improvement. Based on these visual observations, additional model specifications are developed and evaluated. The resulting models are compared using R^2 and MAE, and the model that demonstrates the best performance under the evaluation criteria is selected as the final model.

4 Computational Results

Table 1 Functional Form Selection

No	Variable	Form	R^2	R^2 Difference (vs. Linear)	MAE	MAE Difference (vs. Linear)
0	housing_price	Linear	0.0062		1.9579	
1	housing_price	Log	0.0756	0.0694	1.8624	-0.0955
2	housing_price	Quadratic	0.0444	0.0382	1.9027	-0.0552
3	median_income	Linear	0.2112		1.7129	
4	median_income	Log	0.2128	0.0016	1.6979	-0.015
5	median_income	Quadratic	0.2191	0.0079	1.6972	-0.0157
6	percent_college	Linear	0.2101		1.7169	
7	percent_college	Log	0.2039	-0.0062	1.7241	0.0072
8	percent_college	Quadratic	0.2102	1E-04	1.7169	0
9	percent_married	Linear	0.1962		1.7336	
10	percent_married	Log	0.1692	-0.027	1.7599	0.0263
11	percent_married	Quadratic	0.1965	0.0003	1.7347	0.0011
12	rate_unemployment	Linear	0.1885		1.7406	
13	rate_unemployment	Log	0.1681	-0.0204	1.7736	0.033
14	rate_unemployment	Quadratic	0.1944	0.0059	1.7361	-0.0045

Table 2 Model Compariso

No	Model	Added Variable	Functional Form	R^2	MAE	Change in R^2	Change in MAE
0	Baseline	None	Linear	0.623	1.1598	0	0
1	Extended	rate_unemployment	Quadratic	0.627	1.1528	0.004	-0.0071
2	Extended	housing_price	Log	0.6327	1.1433	0.0097	-0.0165
3	Extended	median_income	Quadratic	0.6323	1.1581	0.0093	-0.0151
4	Extended	percent_married	Linear	0.6242	1.1579	0.0012	-0.0019
5	Extended	percent_college	Linear	0.6332	1.1245	0.0102	-0.0180

Figure 1: Regression Lines and Scatter Plots of Each Variable

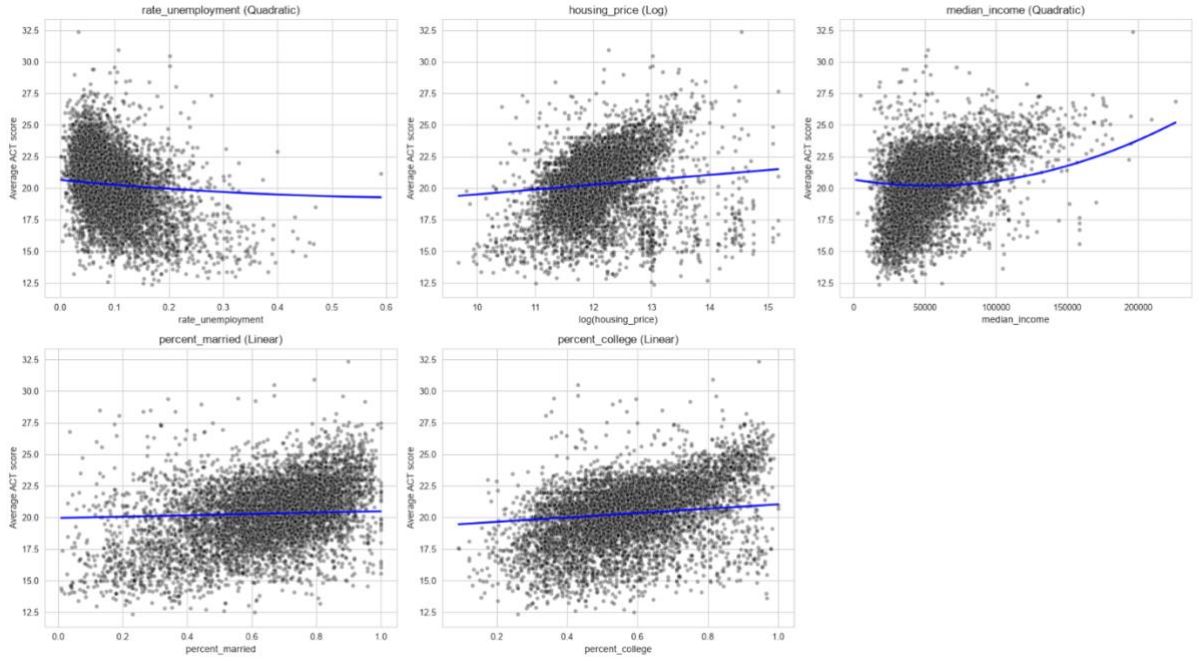


Figure 2: Regression Lines and Residual Plot of Final Model

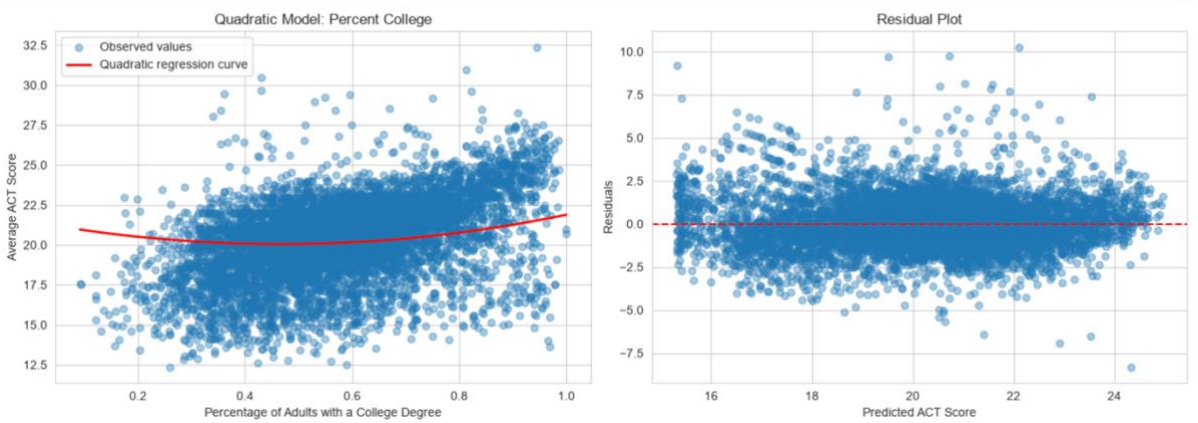


Table 3. OLS Regression Results for the Final Model

Variable	Coefficient	Std. Error	t-value	p-value
Intercept	24.8068	0.204	121.584	< 0.001
percentage_lunch	-8.1035	0.087	-93.02	< 0.001
College attainment	-6.0446	0.647	-9.34	< 0.001
College attainment²	6.4753	0.529	12.242	< 0.001

Table 4. Summary of Final Model Performance

Metric	Value
Observations	7,227
R^2	0.6407
MAE	1.1245

5 Discussion

5.1 Interpretation of Key Findings

Table 1 shows that housing prices perform best with a logarithmic transformation, unemployment and median income with quadratic, and married-couple and college-degree percentages with linear, based on $\boldsymbol{R^2}$ and MAE.

Table 2 indicates that the multiple regression model incorporating the percent lunch variable and the linear term for the percentage of adults with a college degree achieves the best performance among the initial extended models. Although the selected functional form for the percentage of adults with a college degree is linear, Figure 1 shows

a potential nonlinear pattern in the observed data. To investigate whether this pattern can be better captured, a quadratic term for the percentage of adults with a college degree is added to the multiple regression model. The revised model is then evaluated using $\boldsymbol{R^2}$ and MAE to determine whether the additional quadratic term improves model performance.

Table 4 shows that resulting model achieves an $\boldsymbol{R^2}$ of 0.6407 and an MAE of 1.1245, outperforming the baseline simple linear regression model using only the percent_lunch variable. Incorporating the quadratic relationship between college attainment and ACT scores may improve the model's ability to explain variation in school-level ACT scores.

Therefore, by using the result of Table 3 the final mode can be written as below:

$$\boxed{\widehat{ACT} = 24.8068 - 8.1035L - 6.0446C + 6.4753C^2}$$

where (L) represents the percentage of students at the school eligible for free or reduced-price lunch, and (C) represents the percentage of adults with a college degree.

The model can be interpreted as follows:

- Percentage of students eligible for free or reduced-price lunch: Holding the percentage of adults with a college degree constant, a higher percentage of such students is associated with a lower predicted ACT score.
- College attainment: The linear and quadratic terms have coefficients of (-6.0446) and (+6.4753), respectively. This indicates a U-shaped relationship between the percentage of adults with a college degree and predicted ACT scores, holding the percentage of students eligible for free or reduced-price lunch constant.
- $\boldsymbol{R^2}$ = 0.6407, MAE = 1.1245: The model explains approximately 64.07% of the variation in ACT scores in the dataset while the average absolute difference between predicted and actual ACT scores is approximately 1.12 points.
- The residuals (Figure 2) are centered around zero across the range of predicted ACT scores, with no clear curved pattern, which suggests that the quadratic specification captures the main nonlinear relationship reasonably well.

5.2 Limitations and Sources of Error

The dataset includes schools from only 20 states, with most observations concentrated in the eastern United States. Therefore, the findings may not be generalizable to all schools across the United States. In addition, approximately 3% of the housing price values (239 out of 7,986) were imputed, which may have affected the observed relationship between housing prices and ACT scores. Housing prices can vary substantially even between geographically nearby areas, so using neighboring ZIP codes to estimate missing values may introduce measurement error. Future research could improve the imputation process by incorporating additional geographic and socioeconomic information to estimate missing housing prices more accurately.

6 Conclusions

This study examined whether socioeconomic variables improve the prediction of school-level ACT scores beyond a baseline model using the percentage of students eligible for free or reduced-price lunch (FRPL). Among the five additional variables evaluated, the percentage of adults with a college degree achieved the best performance in the initial extended models. Adding a quadratic term for this variable further increased R^2 from 0.6332 to 0.6407, while MAE remained at 1.1245. Compared with the baseline model (R^2 = 0.6230, MAE = 1.1598), the final model explained more variation in ACT scores and reduced prediction error.

However, the dataset covers only 20 states and is concentrated in the eastern United States, limiting the generalizability of the findings. Future research could incorporate more geographically representative data and evaluate model performance on unseen data. Overall, the results suggest that socioeconomic characteristics, particularly adult college attainment, can provide additional information for explaining variation in school-level ACT scores.

References

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